Lesson plan · for the teacher

Training and testing

Data science · Level 3 · Lesson 2

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Split data to train and test.
- Read training and test errors.
- Spot overfitting.

Activity: Hidden cards

1. Warm-up (5 min): Ask: if pupils saw the exact exam questions in advance, would their marks show what they learned?
2. Teach (8 min): Show a wiggly line that hits every dot and a simple line; ask which will do better on new dots.
3. Build (15 min): Groups invent a rule from ten cards and work out its training error.
4. Test (7 min): Reveal the hidden cards; groups work out their test error and compare.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Whose rule did best on the hidden cards?
- Why can't we peek at the test set?
- Which rules overfitted?

Exit ticket

- In one sentence, why do we keep a test set?

Try it at school tomorrow
Groups invent a rule for guessing marks from ten cards, then test it on five hidden cards. Did the rule work as well on the hidden cards?

Practice questions · for you

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1 **Why do we keep 20% of the data away from the model while we build it?**

◯ To make it faster ◯ Because that data is wrong ◯ To check it on pupils it has never seen

2 **A model that scores 100% on its training data will always do well on new data.**

◯ True ◯ False

3 **Keep the first 8 rows for training and the last 2 for testing. Print the size of each set, then the average error of the rule on the test set.**

```
hours = [0, 1, 1, 2, 3, 3, 4, 5, 2, 4]
marks = [8, 10, 11, 12, 13, 15, 16, 18, 15, 13]
train = hours
test = []
print(len(train), len(test))
```

Write your program:

4 **Training error 1, test error 9. What does this suggest?**

◯ Overfitting ◯ A perfect model ◯ The test pupils were easier than the training pupils

5 **Put the steps in order.**

☐ Build the model with the training set

☐ Split the data

☐ Report the test score

☐ Score it once on the test set

✎ **Have you ever trusted a rule because it fitted the past perfectly? How could you have tested it?**

Check your answers

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1 **To check it on pupils it has never seen**

Held-back data gives an honest score.

2 **False**

It may have memorised the training data.

3

```
hours = [0, 1, 1, 2, 3, 3, 4, 5, 2, 4]
marks = [8, 10, 11, 12, 13, 15, 16, 18, 15, 13]
train = hours[:8]
test = hours[8:]
print(len(train), len(test))
err = 0
for i in range(8, 10):
    err = err + abs(marks[i] - (8 + 2 * hours[i]))
print(err / 2)
```

The two test pupils are off by 3 each, so the average test error is 3.0, much worse than the rule does on the training rows.

4 **Overfitting**

A big gap between training and test error is the sign of overfitting.

5 **Top to bottom: 2, 1, 4, 3**

Split first, so the test set stays unseen until the end.